

Christian Muhmann

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🏆 Scholar [Christian Muhmann](#)

Education

- Since 10/25 **PhD in Geodesy and Geoinformation**
German title: Dr.-Ing. (Doctor of Engineering)
Photogrammetry and Robotics Lab, University of Bonn, Bonn (Germany)
- 10/22 – 10/24 **MSc Mechatronics and Robotics, with distinction**
GPA: 1.0 of 5.0 (“excellent”)
Leibniz University Hannover, Hannover (Germany)
Focus: Robotics, Control, and Computational Methods
- 10/21 – 10/24 **MSc Mechanical Engineering, with distinction**
GPA: 1.0 of 5.0 (“excellent”)
Leibniz University Hannover, Hannover (Germany)
Focus: Mechatronic Systems, Robotics, Control, and Computational Methods
- 09/16 – 03/21 **BEng Mechanical Engineering *Duales Studium* (Dual Study Program), with distinction**
GPA: 1.2 of 5.0 (“excellent”)
University of Applied Science Münster, Campus Steinfurt (Germany)

Professional Experience

- Since 10/25 **PhD Candidate**
CARIAD SE, Mönshheim (Germany)
Academic Supervisor: Prof. Dr. Cyrill Stachniss, Photogrammetry and Robotics Lab
Industrial Supervisor: Dr.-Ing. Matthias Zeller, CARIAD SE
• Research topic: Radar-Camera Fusion for Long-Term Localization and Mapping
- 07/25 – 08/25 **Fellow Intern in the Operations Practice (Product Development & Procurement)**
McKinsey & Company, Munich (Germany)
• Received return offer
- 03/21 – 09/21 **Engineer for System Development of Electrical/Electronic (E/E) Architecture**
E/E Systems Engineering, Schmitz Cargobull AG, Münster (Germany)
• Designed and documented E/E components and assemblies for vehicle systems, ensuring system integration and compliance with technical standards; additionally, managed a development project for an electrical signal distribution unit, coordinating cross-functional engineering activities
- 06/19 – 03/21 **Work Term in Duales Studium (Dual Study Program), Part of Bachelor of Engineering**
Schmitz Cargobull AG, Altenberge and Münster (Germany)
• Supported system design, technical documentation, and cross-functional coordination across R&D, product management, and sourcing departments; contributed to process optimization and supported engineering decisions through cost and value analysis
- 08/16 – 06/19 **Apprenticeship as Mechatronics Technician**
Schmitz Cargobull AG, Altenberge (Germany)
• Gained practical experience in prototyping, assembly, wiring, sensor integration, and commissioning of electrical, pneumatic, and hydraulic systems, including basic control systems setup and testing

Research Experience

- 01/24 – 09/24 **International Visiting Graduate Student**
Continuum Robotics Lab, University of Toronto, Toronto (Canada)
Supervisor: Prof. Dr.-Ing. Jessica Burgner-Kahrs, Continuum Robotics Lab
- Master’s thesis: Developed dynamic models for tendon-driven continuum robots (TDCRs) with an arbitrary number of tendons and implemented controllers using a generalized Clarke transformation; validated on a prototype achieving robust trajectory tracking under real-time conditions, leading to a first-author paper accepted at IROS 2025
- 11/22 – 11/23 **Graduate Student Research & Research Assistant**
Institute of Mechatronic Systems, Leibniz University Hannover, Hannover (Germany)
Supervisor: Prof. Dr.-Ing. Thomas Seel, Institute of Mechatronic Systems
- Student research project: Developed recurrent neural network (RNN) models for system dynamics approximation of a pneumatic snake-like soft robot and integrated them into linear and nonlinear MPC frameworks; enabled accurate closed-loop trajectory tracking with an average error of 1.2° in experiments; published as a co-author in IEEE RA-L

Awards & Scholarships

- 01/24 – 09/24 **Scholarship for International Study** from the Dr. Jürgen and Irmgard Ulderup Foundation, awarded for academic excellence to support studies abroad (5,000€)
- 10/22 – 09/24 **German National Scholarship** (*Deutschlandstipendium*) from the Federal Ministry of Education and Research, awarded for talented and high-achieving students (7,200€)
- 10/21 – 09/22 **Dean’s List** from the Leibniz University Hannover, awarded for academic merit
- 03/20 **Karl-Holstein Award** from the North Westphalia Chamber of Industry and Commerce, awarded in recognition of ranking among the top 25 scorers in vocational training exams in North Westphalia (500€)
- 11/19 **State best mechatronics technician in North Rhine-Westphalia** from the South Westphalia Chamber of Commerce and Industry at Hagen, awarded in recognition of ranking among the top scorers in mechatronics technician exams across the state
- 06/19 **“Die Besten” Award** from the North Westphalia Chamber of Industry and Commerce, awarded in recognition of achieving the highest graduation grade (“excellent”) in the mechatronics technician exam

Publications

- Muhmann, C.**, Grassmann, R. M., Bartholdt, M., & Burgner-Kahrs, J. (2025). Toward Dynamic Control of Tendon-driven Continuum Robots using Clarke Transform. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. doi: 10.1109/IROS60139.2025.11247518.
- Schäfke, H., Habich, T.-L., **Muhmann, C.**, Ehlers, S. F. G., Seel, T., & Schappler, M. (2024). Learning-Based Nonlinear Model Predictive Control of Articulated Soft Robots Using Recurrent Neural Networks. *IEEE Robotics and Automation Letters*. doi: 10.1109/LRA.2024.3495579.